

SUBJECT : PROBLEM SOLVING AND TESTING USING JAVA

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Task 2. Access and print the element at a given index in an array.

Task 3. Search for a given element in a sorted array using Binary Search.

Task 4. Find the Maximum element in an array of n integers.

Task 5. Given an array of integers and a positive integer K, write a program to find:
The Kth smallest element.

Task 6. Print all possible pairs of elements from an array of size n.

Task 7. DigitsSum opt: sum of even or odd digits.

Task 8. Nth Fibonacci.

Task 9. Is Palindrome Number?

Task 2. Access and print the element at a given index in an array.

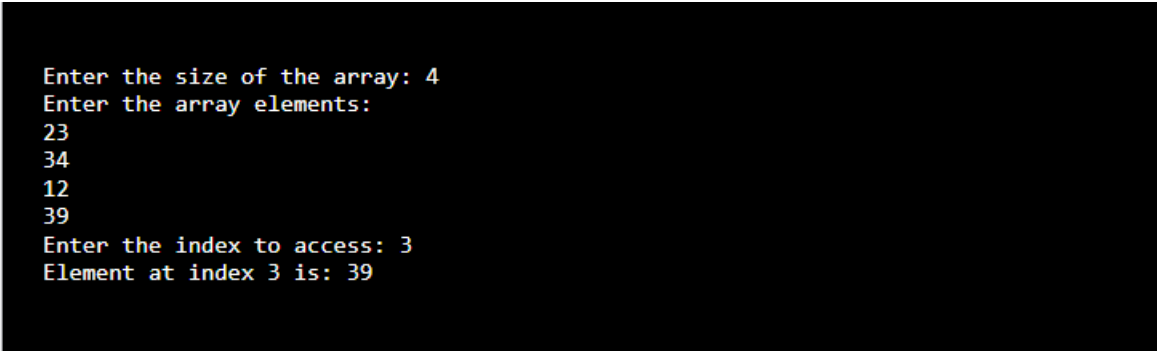
PROGRAM :

```
import java.util.Scanner;
```

```
public class AccessArrayElement {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        // Input array size  
        System.out.print("Enter the size of the array: ");  
        int n = sc.nextInt();  
  
        int[] arr = new int[n];  
  
        // Input array elements  
        System.out.println("Enter the array elements:");  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        // Input index  
        System.out.print("Enter the index to access: ");  
        int index = sc.nextInt();  
  
        // Check if index is valid
```

```
        if (index >= 0 && index < n) {  
            System.out.println("Element at index " + index + " is: " +  
arr[index]);  
        } else {  
            System.out.println("Invalid index!");  
        }  
  
        sc.close();  
    }  
}
```

SAMPLE OUTPUT :



The screenshot shows a terminal window with a black background and white text. The text displays the execution of a Java program. It starts with the prompt 'Enter the size of the array: 4', followed by 'Enter the array elements:'. Then, four numbers are entered on separate lines: 23, 34, 12, and 39. Next, the prompt 'Enter the index to access: 3' is shown, followed by the output 'Element at index 3 is: 39'. At the bottom of the terminal window, there is a status bar with a blue information icon, the text 'Compiled and executed in 29.896 sec(s)', and a small 'x' icon on the right.

```
Enter the size of the array: 4  
Enter the array elements:  
23  
34  
12  
39  
Enter the index to access: 3  
Element at index 3 is: 39
```

 Compiled and executed in 29.896 sec(s) 

Task 3. Search for a given element in a sorted array using Binary Search.

PROGRAM :

```
import java.util.Scanner;
```

```
public class BinarySearchExample {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        // Input array size  
        System.out.print("Enter the size of the array: ");  
        int n = sc.nextInt();  
  
        int[] arr = new int[n];  
  
        // Input sorted array elements  
        System.out.println("Enter the sorted array elements:");  
        for (int i = 0; i < n; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        // Input element to search  
        System.out.print("Enter the element to search: ");  
        int key = sc.nextInt();
```

```
int low = 0;
int high = n - 1;
int foundIndex = -1;

// Binary Search
while (low <= high) {
    int mid = (low + high) / 2;

    if (arr[mid] == key) {
        foundIndex = mid;
        break;
    } else if (arr[mid] < key) {
        low = mid + 1;
    } else {
        high = mid - 1;
    }
}

// Output result
if (foundIndex != -1) {
    System.out.println("Element found at index: " + foundIndex);
} else {
    System.out.println("Element not found.");
}

sc.close();
```

```
}  
}
```

SAMPLE OUTPUT :

```
Enter the size of the array: 3  
Enter the sorted array elements:  
21  
23  
45  
Enter the element to search: 45  
Element found at index: 2
```

 Compiled and executed in 16.429 sec(s)

Task 4. Find the Maximum element in an array of n integers.

PROGRAM :

```
import java.util.Scanner;
```

```
public class MaximumElement {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input array size
```

```
        System.out.print("Enter the size of the array: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        // Input array elements
```

```
        System.out.println("Enter the array elements:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        // Assume first element is maximum
```

```
        int max = arr[0];
```

```
        // Find maximum element
```

```
        for (int i = 1; i < n; i++) {
```

```
        if (arr[i] > max) {  
            max = arr[i];  
        }  
    }  
  
    // Display result  
    System.out.println("Maximum element in the array is: " + max);  
  
    sc.close();  
}  
}
```

SAMPLE OUTPUT :

```
Enter the size of the array: 4  
Enter the array elements:  
25  
30  
24  
12  
Maximum element in the array is: 30  
|
```

 Compiled and executed in 30.902 sec(s)

Task 5. Given an array of integers and a positive integer K, write a program to find: The Kth smallest element.

PROGRAM :

```
import java.util.Arrays;
```

```
import java.util.Scanner;
```

```
public class KthSmallestElement {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input array size
```

```
        System.out.print("Enter the size of the array: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        // Input array elements
```

```
        System.out.println("Enter the array elements:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        // Input K
```

```
        System.out.print("Enter the value of K: ");
```

```
        int k = sc.nextInt();
```

```

// Check if K is valid
if (k <= 0 || k > n) {
    System.out.println("Invalid value of K");
} else {
    // Sort the array
    Arrays.sort(arr);

    // Print the Kth smallest element
    System.out.println("The " + k + "th smallest element is: " + arr[k -
1]);
}

sc.close();
}
}

```

SAMPLE OUTPUT :

Output
Generated files

```

Enter the size of the array: 4
Enter the array elements:
10
30
40
20
Enter the value of K: 2
The 2th smallest element is: 20

```

i Compiled and executed in 13.953 sec(s)

Task 6. Print all possible pairs of elements from an array of size n.

PROGRAM :

```
import java.util.Scanner;
```

```
public class ArrayPairs {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input array size
```

```
        System.out.print("Enter the size of the array: ");
```

```
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];
```

```
        // Input array elements
```

```
        System.out.println("Enter the array elements:");
```

```
        for (int i = 0; i < n; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        // Print all possible pairs
```

```
        System.out.println("All possible pairs are:");
```

```
        for (int i = 0; i < n; i++) {
```

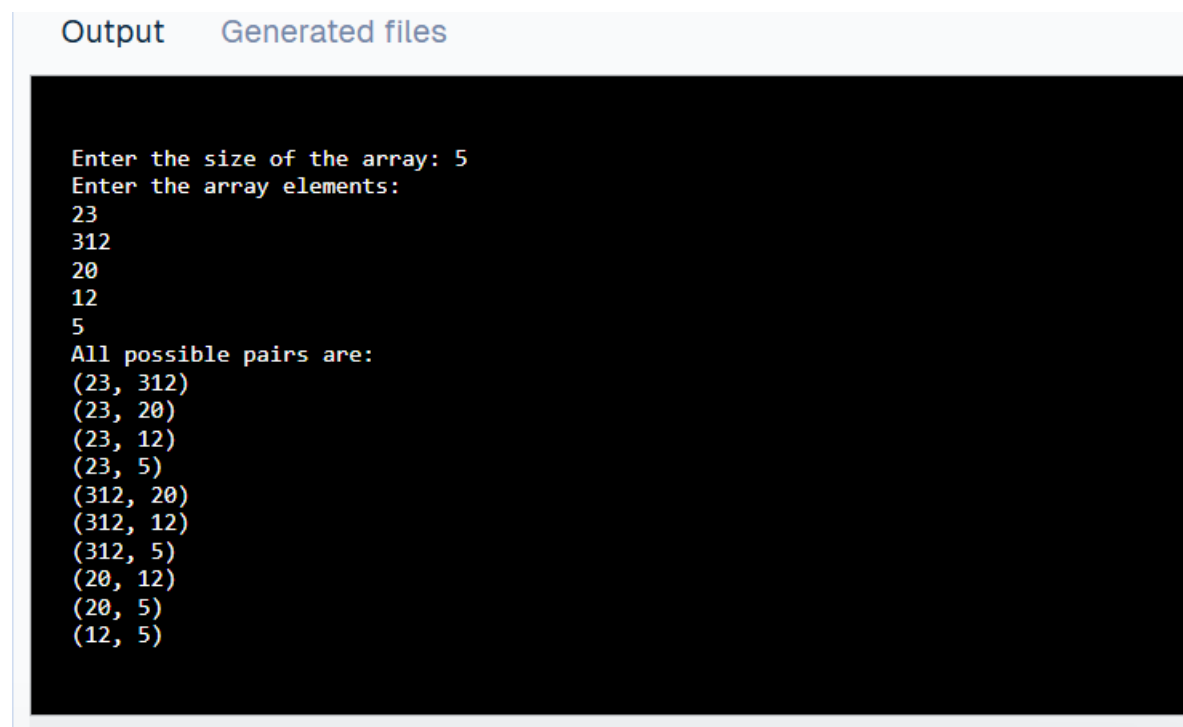
```
            for (int j = i + 1; j < n; j++) {
```

```
                System.out.println("(" + arr[i] + ", " + arr[j] + ")");
```

```
            }
```

```
    }  
  
    sc.close();  
}  
}
```

SAMPLE OUTPUT :



```
Output  Generated files  
  
Enter the size of the array: 5  
Enter the array elements:  
23  
312  
20  
12  
5  
All possible pairs are:  
(23, 312)  
(23, 20)  
(23, 12)  
(23, 5)  
(312, 20)  
(312, 12)  
(312, 5)  
(20, 12)  
(20, 5)  
(12, 5)
```

Task 7. DigitsSum opt: sum of even or odd digits.

PROGRAM :

```
import java.util.Scanner;
```

```
public class DigitsSum {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input number
```

```
        System.out.print("Enter a number: ");
```

```
        int num = sc.nextInt();
```

```
        // Input option
```

```
        System.out.print("Enter option (even/odd): ");
```

```
        String option = sc.next().toLowerCase();
```

```
        int sum = 0;
```

```
        while (num > 0) {
```

```
            int digit = num % 10;
```

```
            if (option.equals("even")) {
```

```
                if (digit % 2 == 0) {
```

```
                    sum += digit;
```

```
                }
```

```
    } else if (option.equals("odd")) {
        if (digit % 2 != 0) {
            sum += digit;
        }
    }

    num = num / 10;
}

if (option.equals("even")) {
    System.out.println("Sum of even digits = " + sum);
} else if (option.equals("odd")) {
    System.out.println("Sum of odd digits = " + sum);
} else {
    System.out.println("Invalid option!");
}

sc.close();
}
}
```

SAMPLE OUTPUT :

Output Generated files

```
Enter a number: 546789
Enter option (even/odd): even
Sum of even digits = 18
|
```

 Compiled and executed in **15.507 sec(s)**

Task 8. Nth Fibonacci.

PROGRAM :

```
import java.util.Scanner;
```

```
public class NthFibonacci {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input the value of N
```

```
        System.out.print("Enter the value of N: ");
```

```
        int n = sc.nextInt();
```

```
        int first = 0, second = 1, next = 0;
```

```
        if (n == 1) {
```

```
            System.out.println("Nth Fibonacci number is: " + first);
```

```
        } else if (n == 2) {
```

```
            System.out.println("Nth Fibonacci number is: " + second);
```

```
        } else {
```

```
            for (int i = 3; i <= n; i++) {
```

```
                next = first + second;
```

```
                first = second;
```

```
                second = next;
```

```
            }
```

```
            System.out.println("Nth Fibonacci number is: " + next);
```

```
    }  
  
    sc.close();  
}  
}
```

SAMPLE OUTPUT :

Output Generated files

```
Enter the value of N: 7  
Nth Fibonacci number is: 8  
|
```

 Compiled and executed in 34.421 sec(s)

Task 9. Is Palindrome Number?

PROGRAM :

```
import java.util.Scanner;
```

```
public class PalindromeNumber {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        // Input the number
```

```
        System.out.print("Enter a number: ");
```

```
        int num = sc.nextInt();
```

```
        int original = num;
```

```
        int reverse = 0;
```

```
        // Reverse the number
```

```
        while (num > 0) {
```

```
            int digit = num % 10;
```

```
            reverse = reverse * 10 + digit;
```

```
            num = num / 10;
```

```
        }
```

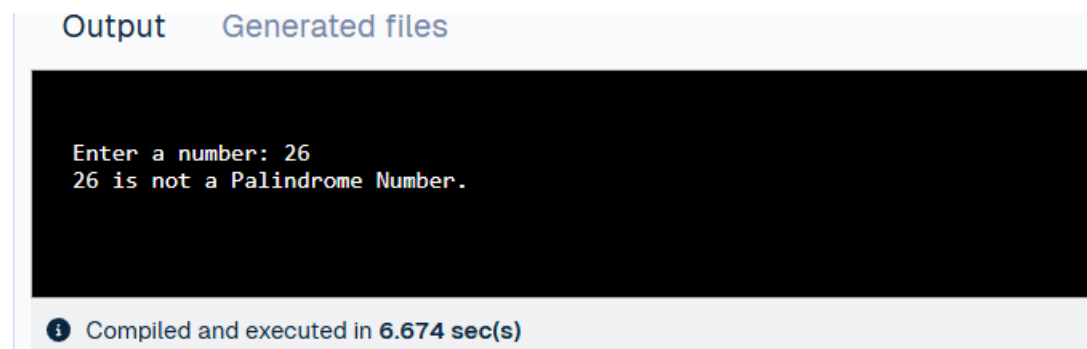
```
        // Check palindrome
```

```
        if (original == reverse) {
```

```
            System.out.println(original + " is a Palindrome Number.");
```

```
    } else {  
        System.out.println(original + " is not a Palindrome Number.");  
    }  
  
    sc.close();  
}  
}
```

SAMPLE OUTPUT :



The screenshot shows a code execution interface with two tabs: "Output" and "Generated files". The "Output" tab is active, displaying the following text on a black background: "Enter a number: 26" and "26 is not a Palindrome Number.". Below the output, a status bar indicates "Compiled and executed in 6.674 sec(s)".